

Fundamental parameters of Ap-stars GO And, 84 UMa and κ Psc.

A. M. Romanovskaya^{*1}, D. V. Shulyak², T. A. Ryabchikova¹, and T. M. Sitnova¹

¹ Institute of Astronomy, Russian Academy of Sciences, Pyatnitskaya 48, 119017, Moscow, Russia

² Instituto de Astrofísica de Andalucía - CSIC, c/ Glorieta de la Astronomía s/n, 18008 Granada, Spain.

Received April 00, 000; accepted March 00, 0000

ABSTRACT

Aims. The aim of this work is to carry out the determination of fundamental parameters of three Ap stars: GO And (HD 4778), κ Psc (HD 220825), and 84 UMa (HD 120198) using spectroscopic techniques. By analysing these stars we complete the sample of Ap stars for which fundamental parameters had been additionally derived by means of interferometry, thus making it possible to cross compare results derived by direct and indirect methods.

Methods. Our study is based on the analysis of high-resolution high S/N spectra obtained with ESPaDOnS spectrograph. We used iterative method of fundamental parameters determinations which includes self-consistent modelling of stellar atmosphere taking into account individual abundances of chemical elements and subsequent fitting theoretical spectral energy distribution (SED) to the observed one. The quality of spectroscopic determinations was evaluated through a comparison with the interferometric results.

Results. For all investigated stars we determined fundamental parameters and derived chemical abundances that turn to be typical for Ap stars and are characterized mainly by gradual increase of heavy elements atmospheric abundance from an order of magnitude for iron peak elements up to very significant excesses of 3–4 dex of the rare-earth elements relative to the solar values. The only exception is Ba which abundance is close to the solar one. There is also a significant He deficiency in the atmospheres of HD 120198 and HD 220825, whereas He abundance in HD 4778 is close to the solar one. We do not find significant Fe and Cr stratification. Using these abundances we constructed self-consistent atmospheric models for each star. The effect of the surface chemical inhomogeneity on the derived fundamental parameters did not exceed ± 100 K in effective temperature which lies within a range of errors in similar self-consistent analyses of Ap stars. Finally, we compared spectroscopically derived effective temperatures, radii, and luminosity for 13 out of 14 Ap stars in benchmarking sample with the interferometric results. While radii and luminosity agree within the quoted errors of both determinations, spectroscopic effective temperatures are higher than the interferometric ones for stars with $T_{\text{eff}} > 9000$ K. The observed hydrogen line profiles favor the spectroscopically derived temperatures.

Key words. chemically peculiar stars – fundamental parameters – abundances

1. Introduction

Magnetic chemically peculiar (Ap) stars belong to the Main Sequence (MS) group between spectral types B5 to F5. These stars have global magnetic fields with the measured longitudinal components varying from few tens of Gauss (Aurière et al. 2007) to tens of kilo-Gauss (see catalogue Romanyuk & Kudryavtsev 2008). The magnetic fields have predominantly poloidal structure. These stars are also slow rotators with the rotation velocities less than 100 km s^{-1} , supposed to be due to magnetic braking (Mathys 2017). Ap stars have the same masses and luminosity as the normal MS stars of similar temperatures.

The presence of strong magnetic fields significantly affects the atmospheric chemical composition in Ap-stars. The observed abundance anomalies were explained by Michaud (Michaud 1970) as a result of particle diffusion.

Studies of the stratification of chemical elements by spectral observations showed that most of the elements have vertical abundance gradients (Shulyak et al. 2009). For Ap stars, the standard photometric and spectroscopic calibrations developed to determine the fundamental parameters of normal stars are often inapplicable. In magnetic stars, the presence of magnetic fields and significant individual chemical composition anomalies re-

quires a detailed study of stellar chemistry in order to construct an adequate model atmosphere that can predict the observed flux distribution. A self-consistent procedure for the simultaneous modeling of the spectrum and flux distribution was first proposed by Shulyak et al. (2009) for the Ap star HD 24712. Then, this method was used to determine the fundamental parameters of several other Ap stars (Kochukhov et al. 2009; Shulyak et al. 2010b; Pandey et al. 2011; Shulyak et al. 2013; Nesvácil et al. 2013).

The accuracy of fundamental parameters determination by spectroscopic methods can be estimated by their comparison with the results of direct determinations of stellar radii by means of interferometric observations. Combining the latter with (spectro)photometric absolute calibrated flux one can estimate stellar temperature and luminosity for a known distance. Interferometry provides a model independent way to estimate stellar radii which can serve as a reference for testing theoretical models. Recently, Perraut et al. (2020) presented interferometric observations of a sample of Ap stars obtained with CHARA array¹ (ten Brummelaar et al. 2010) and starting from 2008 (see their Table 1). The sample contains stars of late B – early F spectral types and diverse stellar ages so that the atmospheric properties of these stars such as, e.g., magnetic field strength and chemical

^{*} annarom@inasan.ru

¹ <http://www.chara.gsu.edu/>

abundances, can substantially differ from star to star. Note that the sample was primarily chosen according to instrument limitations, that required stars to be bright ($V^m < 6$) and with relatively high angular diameters > 0.2 mas. Given an unique opportunity to test our theoretical models and analysis methods against independent approaches, we initiated a follow-up study of interferometric targets by means of spectroscopy. Our early results indicated a good agreement between two methods (see, e.g. Romanovskaya et al. 2019). The aim of this paper is to complete the follow-up spectroscopic program and present the results of detailed spectroscopic analysis of last three Ap stars from the sample of interferometric measurements: HD 4778 (GO And), HD 120198 (84 UMa), and HD 220825 (κ Psc). We also present a final comparison between the results obtained by direct (interferometry) and indirect (spectroscopy) methods.

2. Observations

Spectroscopic observations were extracted from the ESPaDOnS spectrograph (Canada-France-Hawaii Telescope) archive² and cover the spectral range of 3700–10 500 Å. The resolving power of the instrument is $R = \lambda/\Delta\lambda = 68\,000$.

Photometric/spectrophotometric observations in different spectral regions were employed to construct the spectral energy distribution for each star. For UV region we used observations obtained with S2/68 telescope of TD1 mission (European Space Research Organization (ESRO) satellite), which measured absolute flux in four bands: 1565 Å, 1965 Å, 2365 Å, and 2740 Å (Thompson et al. 1978), and data from IUE (International Ultraviolet Explorer)³. In the optical range we used data from Adelman’s spectrophotometric catalogue (Adelman et al. 1989) and/or Johnson photometry in V filter taken from ADS SIMBAD online database⁴. Photometric data in IR region were taken from the 2MASS catalogue (2Micron All-Sky Survey Cutri et al. (2003)), which contains an overview of the entire sky in J (1.25 microns), H (1.65 microns), and Ks (2.17 microns) filters. Observations were transformed into absolute flux using calibrations given in Cohen et al. (2003).

- HD 4778: UV data is available in both IUE and TD1 archives; in optical range there is no Adelman’s spectrophotometry and thus we used only Johnson photometry, as well as 2MASS photometric measurements in IR.
- HD 120198: photometric and spectrophotometric data is available from IUE, TD1, Adelman’s, and 2MASS catalogues.
- HD 220825: IUE and TD1 data is missing, and thus we use only Adelman’s and 2MASS measurements to build spectral energy distribution.

3. Spectroscopic analysis

3.1. Methods

The fundamental parameters of the stars were obtained by various methods in previous works (Table 1). Here and throughout the paper an error of the measurement in the last digits is given in parentheses. The latest parameter determinations were performed by interferometry (Perraut et al. 2020). Kochukhov &

Bagnulo (2006) estimated the effective temperatures and luminosity of stars through the calibrations of the Geneva photometric system using the revised Hipparcos parallaxes van Leeuwen (2007). The radii of the stars were calculated using the Stefan-Boltzmann formula. Hubrig et al. (2007) determined the effective temperatures from Geneva system and Strömgren system photometry, and the luminosity have been obtained by taking into account the bolometric corrections. Lipski & Stępień (2008) determined T_{eff} from fitting theoretical flux to the observed spectral energy distribution; $\log g$ in this work was assumed to be $\log g = 4.0$. Adelman & Rayle (2000) derived fundamental parameters through different methods. Netopil et al. (2008) collected fundamental parameters from different sources; the values of T_{eff} given in Table 1 were obtained using infrared flux method.

Here, we used iterative method of the determination of stellar fundamental parameters (for more details, see Shulyak et al. 2009). It includes abundance and stratification calculations, if necessary, and the subsequent refinement of the atmospheric model by fitting calculated flux to the observed spectral energy distribution, as well as comparing observed and predicted hydrogen line profiles. It means that abundance analysis and parameters determinations are interconnected. This allows us to obtain fairly accurate atmospheric parameters, normally within a few iterations.

To calculate the model atmosphere it is necessary to know the basic parameters of the star – the effective temperature T_{eff} , the logarithm of surface gravity $\log g$, metallicity, and also the magnetic field strength. We took initial atmospheric parameters from the work by Kochukhov & Bagnulo (2006). The initial models atmosphere were calculated using the ATLAS9 program (Kurucz 1993). In this model the absorption in spectral lines is taken into account statistically through the opacity distribution function (ODF) as part of the overall opacity. ATLAS9 model was used for preliminary abundance analysis in chemically homogeneous stellar atmospheres. These calculations were carried out using spectral synthesis program SYNMAST (for a detailed description see Kochukhov 2007; Kochukhov et al. 2010) that takes into account the magnetic field in radiative transfer calculation.

Atomic parameters of spectral lines were taken from the 3rd version of Vienna Atomic Line Database VALD (Kupka et al. 1999; Ryabchikova et al. 2015).

Atmospheric parameters were refined by comparing the observed and theoretical spectral energy distribution (SED) calibrated to absolute flux units. For this and following steps of the iterative procedure we used LLMODELS code for model atmosphere calculations (Shulyak et al. 2004) which accounts for individual atmospheric abundances in opacity calculation.

3.1.1. Magnetic field and rotation velocity

To determine the element abundance, we calculated the synthetic spectrum and fitted it to the observed one using the visualization program BINMAG6 (Kochukhov 2018).

The magnetic fields of our program stars are strong enough to play a significant role in the formation of spectral lines and therefore should be taken into account in spectrum synthesis. Due to a rather large rotation velocity $v \sin i$ it was not possible to estimate surface magnetic field modulus $\langle B_s \rangle$ from Zeeman splitting or from magnetic differential broadening (Romanovskaya et al. 2020a) for all stars. Therefore, we took $\langle B_s \rangle$ values from the literature: 2.6 kG for HD 4778 (Glagolevskij 2019); 1.6 kG for HD 120198, and 2.0 kG for HD 220825 (Kochukhov 2017).

² <http://www.cadc-ccda.hia-ihp.nrc-cnrc.gc.ca/en/cfht/>

³ <http://archive.stsci.edu/iue/>

⁴ <http://simbad.u-strasbg.fr/simbad/>

Table 1. Literature data for the fundamental parameters of program stars. Several studies do not provide uncertainties of measurements.

HD	T_{eff}	$\log g$	$\log(L/L_{\odot})$	$R/R_{\odot}$	Ref.	Method
4778	9977(300)		1.51(07)	1.91(04)	Kochukhov & Bagnulo (2006)	Photometry
	9135(400)		1.54(05)	2.36(12)	Perraut et al. (2020)	Interferometry
	9375(300)	4.14(09)	1.48(07)	2.10(22)	Hubrig et al. (2007)	Photometry
120198	10543(390)		1.65(06)	2.01(01)	Kochukhov & Bagnulo (2006)	Photometry
	9865(370)		1.65(04)	2.28(10)	Perraut et al. (2020)	Interferometry
	9750	4.00		2.22	Lipski & Stepień (2008)	SED
	10500	4.19			Adelman & Rayle (2000)	Spectrophotometry
	11234	4.07			Adelman & Rayle (2000)	<i>ubvyβ</i>
	10400	4.00			Adelman & Rayle (2000)	H_{γ}
220825	9078(290)		1.35(04)	1.92(03)	Kochukhov & Bagnulo (2006)	Photometry
	8790(230)		1.24(04)	1.78(03)	Perraut et al. (2020)	Interferometry
	9200(80)				Netopil et al. (2008)	IRFM
	9180(370)	4.27(07)	1.28(03)	1.73(14)	Sikora et al. (2019)	SED
	9080(290)	4.16(08)	1.383(53)	2.00(18)	Hubrig et al. (2007)	Photometry
	9250	3.75			Ryabchikova et al. (1996)	IRFM

These values of the magnetic field were employed in all synthetic spectrum calculations.

We derived $v_e \sin i$ from several line blends with different sensitivity to the magnetic field. It was done by utilizing the fitting procedure implemented in the `BinMag6` code. The final values are the following: $v_e \sin i = 31.6 \pm 3.8 \text{ km s}^{-1}$ (HD 4778), $v_e \sin i = 56.0 \pm 0.5 \text{ km s}^{-1}$ (HD 120198), and $v_e \sin i = 38.4 \pm 1.5 \text{ km s}^{-1}$ (HD 220825).

3.1.2. Chemical abundances

The average abundance of chemical elements was determined by using two methods. First, we determined the abundance through the measured equivalent widths of spectral lines by using the `WIDSYN` code (Shulyak et al. 2013). `WIDSYN` calculates theoretical equivalent widths by taking into account polarization in spectral lines. We assumed surface magnetic field to be constant with atmospheric depth and defined by the radial component of the magnetic field vector $\langle B_r \rangle$. This method is fast, but was not always useful because of a rather high rotation velocity $v_e \sin i$ of stars and rich line spectrum that makes lines of different elements to blend together and affect measurements of their equivalent widths. Therefore, in order to obtain more accurate values, we used the method of fitting the synthetic line profiles to the observed ones. In this method, the following parameters must be optimized: rotation velocity along the line of sight $v_e \sin i$, radial velocity V_r , radial component of the magnetic field B_r and, if necessary, the meridional B_m component of the magnetic field, and finally the element abundance. Fitting the theoretical line profiles to the observed ones was carried out using the `BinMag6` code along with the `SYNMAST` code (Kochukhov 2007). Here and below the abundance is given as the logarithm of the ratio of the number of given element atoms to the total number of atoms of all elements $\log(N_{\text{el}}/N_{\text{tot}})$. Examples of comparison between two methods of abundance determination are presented in Table 2 for the HD 220825. It is seen that the equivalent widths method results in systematically higher abundances compared to the direct line profile fitting. This is because of the line profile distortion where the lines of different elements blend together with the spectral line of an element whose abundance we want to measure. This artificially increases the width of the spectral line under investigation, and a higher abundance is needed to reproduce the measured equivalent width. To the contrary, the direct spectrum synthesis accounts for line blending and distortion of shapes of spectral lines self-consistently. Therefore, we

finally adopted the abundance values derived using the method of spectrum synthesis.

Table 2. Comparison of several elemental abundances in I and II ionization stages, calculated through equivalent widths (EW) and spectrum synthesis (SS) for HD 220825.

Ion	$\log \left(\frac{N}{N_{\text{tot}}} \right)_{\text{EW}}$	$\log \left(\frac{N}{N_{\text{tot}}} \right)_{\text{SS}}$
Mg I	-3.64(24)	-3.92(23)
Mg II	-3.84:	-4.45:
Fe I	-3.23(40)	-3.37(23)
Fe II	-3.08(34)	-3.58(30)
Mn II	-4.38(25)	-4.82(05)
Ti II	-5.03(31)	-5.91(31)
Cr I	-3.57(36)	-3.79(24)
Cr II	-3.55(25)	-3.84(36)
Nd III	-6.66(61)	-6.82(25)
Eu II	-7.51(61)	-8.38(03)
Eu III	-6.37:	-6.40:

For elements up to Ba the abundances were obtained from the lines of neutral and first ions while for the rare earth elements (REE) we used the lines of the first and second ions. For all elements but O, the abundances were obtained in the local thermodynamic equilibrium (LTE) approximation. The oxygen abundance was determined from the lines of the infrared triplet O I 7771–7775 Å taking into account the deviation from LTE (NLTE). We adopted O I model atom of Przybilla et al. (2000), which was updated by including accurate data for collisions with electrons and hydrogen atoms as described in Sitnova et al. (2013) and Sitnova & Mashonkina (2018), respectively. The isotopic structure of Ba II and Eu II was taken into account, but we neglected hyperfine splitting (HFS) due to the complex interaction between hyperfine and magnetic splitting (Landi Degl’Innocenti 1975). Therefore we caution that the abundance of elements with odd isotopes (Mn, Pr, Eu) might be overestimated.

For HD 120198 it was impossible to derive accurate Si abundance from the spectrum synthesis due to a lack of suitable unblended lines. However we noticed that a change in the Si abundance noticeably affects the part of the flux in the ultraviolet spectral region (see Fig. 1). Decreasing the Si abundance by 1.0 dex leads to an overestimated flux in the region of 1500 Å compared to the observed data from the IUE and TD1. At the same time, in other regions of the spectrum the fit remained un-

Table 3. The mean values of chemical abundances $\log(N_{el}/N_{tot})$, calculated from equivalent widths and spectrum synthesis of N lines for model atmospheres 9605g40 (HD 4778), 10174g42 (HD 120198), and 9470g42 (HD 220825). Values derived from single line are marked with a colon. The standard deviation is given in parentheses, and was assumed to be 0.5 dex in cases when the measurement was done from a single line only. The last column contains solar photospheric abundances that were taken from Asplund et al. (2021).

Ion	Abundance, $\log(N_{el}/N_{tot})$					
	HD 4778	N	HD 120198	N	HD 220825	N
C I	-3.80(50)	1	-3.31(50)	3	-4.26(15)	2
N I	-4.63(50)	1			-5.02(50)	1
O I	-4.84(08)	3	-5.21(07)	3	-5.33(07)	3
Na I	-5.91(50)	1			-4.81(50)	1
Mg I	-4.77(56)	4	-4.37(41)	3	-3.92(23)	4
Mg II	-4.78(28)	3	-4.19(24)	2	-4.45(50)	1
Al I					-4.14(50)	1
Al II	-4.76(29)	2	-5.19(01)	2	-4.67(50)	1
Si I			-3.84(10)	2	-4.19(50)	1
Si II	-4.67(17)	3	-4.08(12)	3	-5.72(38)	3
Ca I	-4.08(01)	2	-5.60(50)	1	-4.99(50)	1
Ca II	-4.80(20)	2	-6.37(07)	2	-5.71(50)	1
Ti II	-5.95(25)	5	-6.34(54)	4	-5.91(31)	7
Cr I	-3.73(36)	11	-3.38(16)	10	-3.79(24)	11
Cr II	-3.75(34)	24	-3.36(20)	11	-3.84(36)	31
Mn I	-4.95(50)	1			-5.36(50)	1
Mn II	-4.56(29)	11	-5.75(50)	1	-4.82(05)	2
Fe I	-2.98(27)	16	-3.26(23)	5	-3.37(23)	22
Fe II	-3.04(25)	38	-3.36(25)	14	-3.58(30)	33
Sr I	-5.57(50)	1			-5.47(50)	1
Sr II	-7.10(50)	1	-7.51(50)	1	-7.54(50)	1
Y II	-7.61(50)	1				
Zr II	-8.53(50)	1				
Ba II	-9.82(50)	1	-9.50(50)	1	-8.41(50)	1
La II	-7.35(13)	2	-8.17(50)	1	-8.50(50)	1
Ce II	-6.99(31)	7	-7.29(50)	1	-7.45(50)	1
Pr II	-7.86(26)	6	-8.34(05)	2	-7.26(23)	3
Nd II	-7.72(50)	1			-8.20(50)	1
Nd III	-7.56(35)	7	-7.94(43)	9	-6.82(25)	5
Sm II	-7.63(10)	3	-7.75(50)	1	-7.80(50)	1
Eu II	-7.31(16)	4	-7.65(41)	4	-8.38(03)	2
Eu III	-5.47(50)	1	-6.60(50)	1	-6.37(50)	1
Gd II	-8.22(50)	1				
Lu II	-8.47(30)	2			-7.77(50)	1

changed with only a slight increase in the effective temperature by about 50 K. It means that the flux observations in UV spectral region may serve as an important indicators of Si abundance in the atmospheres of some Ap stars, similar to what was observed in hotter B-type chemically peculiar stars (see, e.g., Krtićka et al. 2012).

The final abundances in the atmospheres of the program stars together with the number of lines used in the analysis are collected in Table 3 and are shown in Fig. 2 relative to the solar values.

In summary, the abundances were estimated for 24 elements. For 7 elements abundances were derived from the lines of two consecutive ions. In all three stars we obtained typical abundance trends observed for Ap stars of SrCrEu type: deficiency of C, N, and O; close to solar abundance of Na, Mg, Ca; large Cr, Sr, and REE overabundance in agreement with previous works (see, for example, Shulyak et al. (2010b); Romanovskaya et al. (2019)). A deficit of He is observed in the stars HD 120198 and HD 220825, respectively, since He I 4471 Å line is practically not visible in their spectra. Therefore, He-weak models of atmospheres were calculated with He abundance of $\log(\text{He}/N_{tot}) = -4.05$ and, accordingly, with an $\text{H}/N_{tot} = 0.99$. A similar He deficiency was observed for, e.g., another Ap star HD 108662 (Romanovskaya et al. 2020b). In the atmosphere of the star HD 4778 He I is observed and we used the solar abundance $\log(\text{He}/N_{tot}) = -1.05$ and $\text{H}/N_{tot} = -0.91$, respectively.

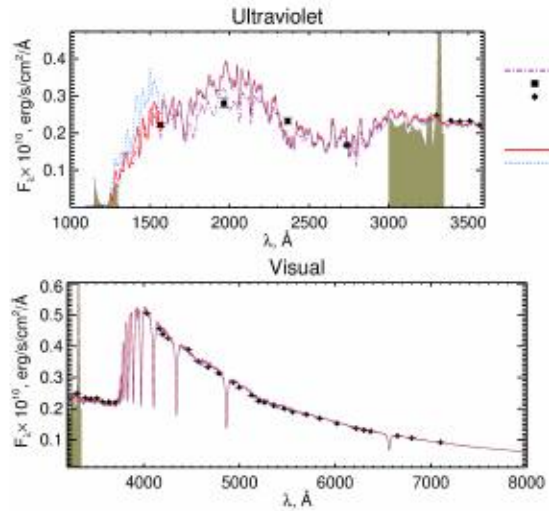


Fig. 1. SED of HD 120198 in UV and visual spectral regions. Red and blue-dotted lines show theoretical SED calculations with the same final model atmosphere parameters t10174g42 but with different Si abundance: $\log(N_{Si}/N_{tot}) = -3.82$ and -4.81 , respectively. Shaded parts indicate spectral regions excluded from the fit.

The ionization equilibrium is conserved, within the error bars, for Mg, Ca, Cr, Mn, and Fe, that gives evidence for the absence of significant vertical stratification of these elements. We consider possible effect of Fe stratification on flux distribution later, in Section 3.1.3. The only rare-earth element with observable lines of two ions (first and second) is Eu which shows typical REE-anomaly observed in most Ap stars in the effective temperature range of 7000 K – 10 000 K (Ryabchikova & Romanovskaya 2017). The most interesting feature in abundance distribution is the abundance sequence Sr-Y-Zr-Ba-REE, where one observes a sharp decrease of Ba abundance up to the solar value and ≈ 3 dex overabundance of the neighboring elements Sr and REEs. Overabundances (in less extent) of heavy elements Sr-Zr-Ba-Nd were also derived in normal A-stars and in Am stars (Mashonkina et al. 2020), but there Ba had a largest overabundance among neighbor elements Sr, Zr, and Nd. This different behaviour of Ba certainly indicates a difference in the mechanisms that control the atmospheric abundance distribution in normal and peculiar A-type stars.

3.1.3. Stratification analysis

The vertical distribution of chemical elements in the stellar atmosphere (stratification profile) depends on the effective temperature T_{eff} , surface gravity $\log g$ and magnetic field strength $\langle B_s \rangle$ of the star (LeBlanc et al. 2009; Alecian & Stift 2010). Theoretical diffusion calculations of some elements provided by Babel (1992) and later by (LeBlanc et al. 2009) showed that, as a first approximation, the stratification profile of an element can be represented by a step function with four parameters: the element abundance in the upper atmosphere, the element abundance in the lower atmosphere, the location of the abundance jump, and the width of this jump. This was then widely used in the analysis of spectral observations (e.g. Wade et al. (2001); Ryabchikova et al. (2005)).

The accuracy of the stratification study strongly depends on the choice of spectral lines. To study the stratification in the atmospheres of our sample stars, a set of most suitable lines must be chosen. It includes individual and preferably unblended

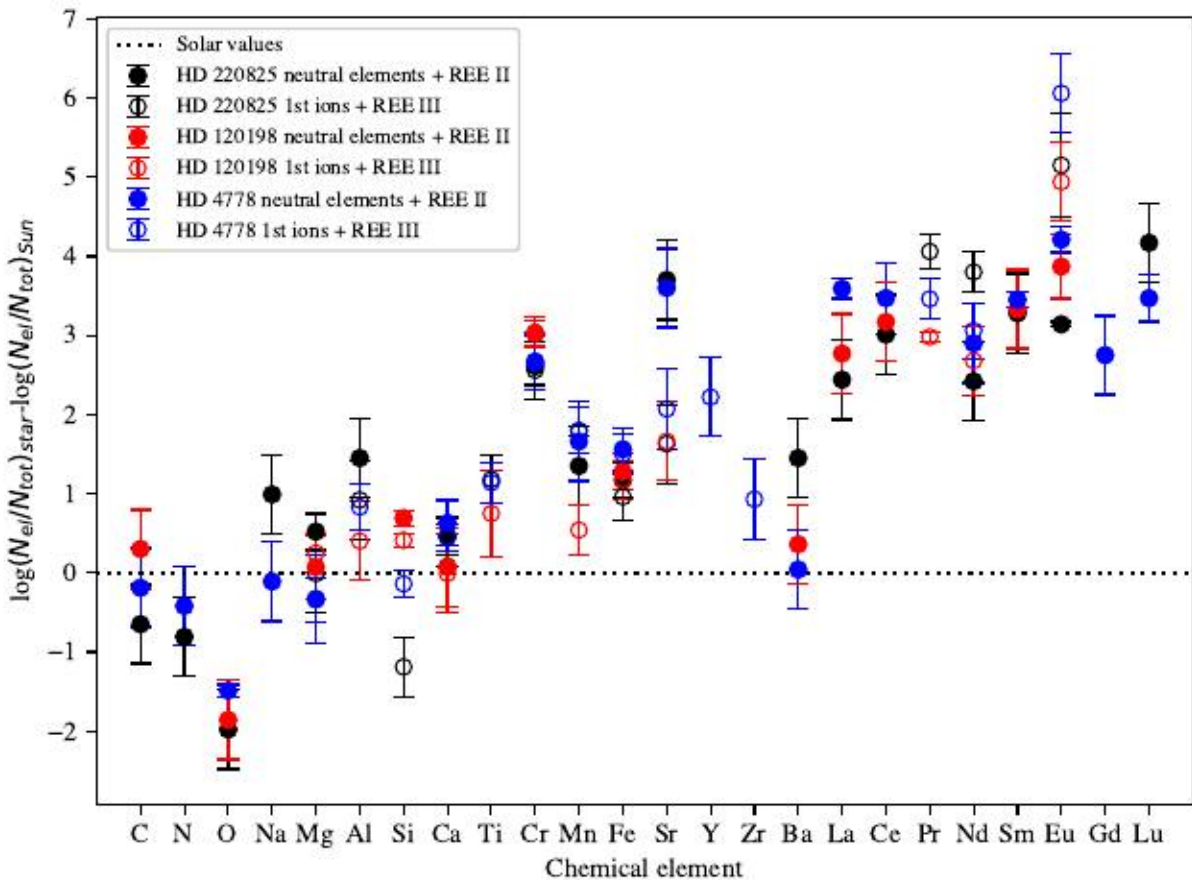


Fig. 2. Atmospheric abundances in HD 4778 (blue circles), HD 120198 (red circles) and HD 220825 (black circles) relative to the solar values.

lines with different excitation potentials E_i , different intensities/equivalent widths EW , i.e. lines that are formed at different optical depths and thus probe different atmospheric layers.

In atmospheres of Ap stars, the elements Fe and Cr are the best elements for the stratification analysis due to a large number of spectral lines available.

A strong abundance gradient of these elements is observed mainly in cool stars (6000 K – 8000 K), where the abundance jump amplitude is decreasing with increasing T_{eff} and shifts to the upper atmospheric layers (LeBlanc et al. 2009). The stars in our sample have rather high effective temperatures and high rotational velocities $v \sin i$, which causes the spectral lines to be broadened and blended. Because of this, it was difficult to perform accurate stratification analysis. Nevertheless, in order to estimate a possible presence of element stratification in our sample stars, we tried two sets of Fe lines (with a difference in two lines) for the analysis of Fe stratification in HD 220825. The resulting stratification profiles were derived using the code DDAFrr (Ryabchikova et al. 2005) and are shown in Fig. 3a. It can be seen that the stratification profiles are very different even though the underlying models are almost similar: the difference in 26 K in their T_{eff} is obviously too small to cause such a variation in the stratification profiles. The first set of lines results in very much homogeneous Fe distribution (red curve), while using the second set of Fe lines we obtained a solution with large uncertainties for the abundance at low and upper atmospheric boundaries (blue curve). This second solution also resulted in a worse fit to hydrogen Balmer lines, as shown in Fig. 3b which compares observed

and predicted $H\beta$ line profiles calculated assuming stratified and non-stratified models, respectively. Similar to the case of SED fitting (see next Section), we found out that the chemically homogeneous model atmosphere describes the observed $H\beta$ line profile better compared to a stratified one. Therefore, we finally chose models without stratification for further analysis for our sample stars.

4. Determination of the fundamental parameters

Our determination of fundamental parameters of stars is based on fitting theoretical spectral energy distribution to the observed one.

We calculated the flux emerged from the surface of a star using the LLMODELS code. The distances to the stars are defined by stellar parallax values that we take from GAIA DR2 catalog (Gaia Collaboration 2018). We also took into account an interstellar absorption. HD 4778 is located at the distance of 107.268 pc from the Sun and the correction for the interstellar reddening $A_v = 0.065$ ($E(B - V) = 0.02$) was applied (Amôres & Lépine 2005). Other stars, HD 120198 and HD 220825, are located closer to the Sun which results in weaker interstellar absorption: $E(B - V) = 0.002(14)$ for HD 220825 and $E(B - V) = 0.008(11)$ for HD 120198, respectively (Lallement et al. 2014). Therefore we neglected the corrections for interstellar absorption for these two stars.

We compared the theoretical flux calculated with our model atmosphere code with the observed SED. For a given abun-

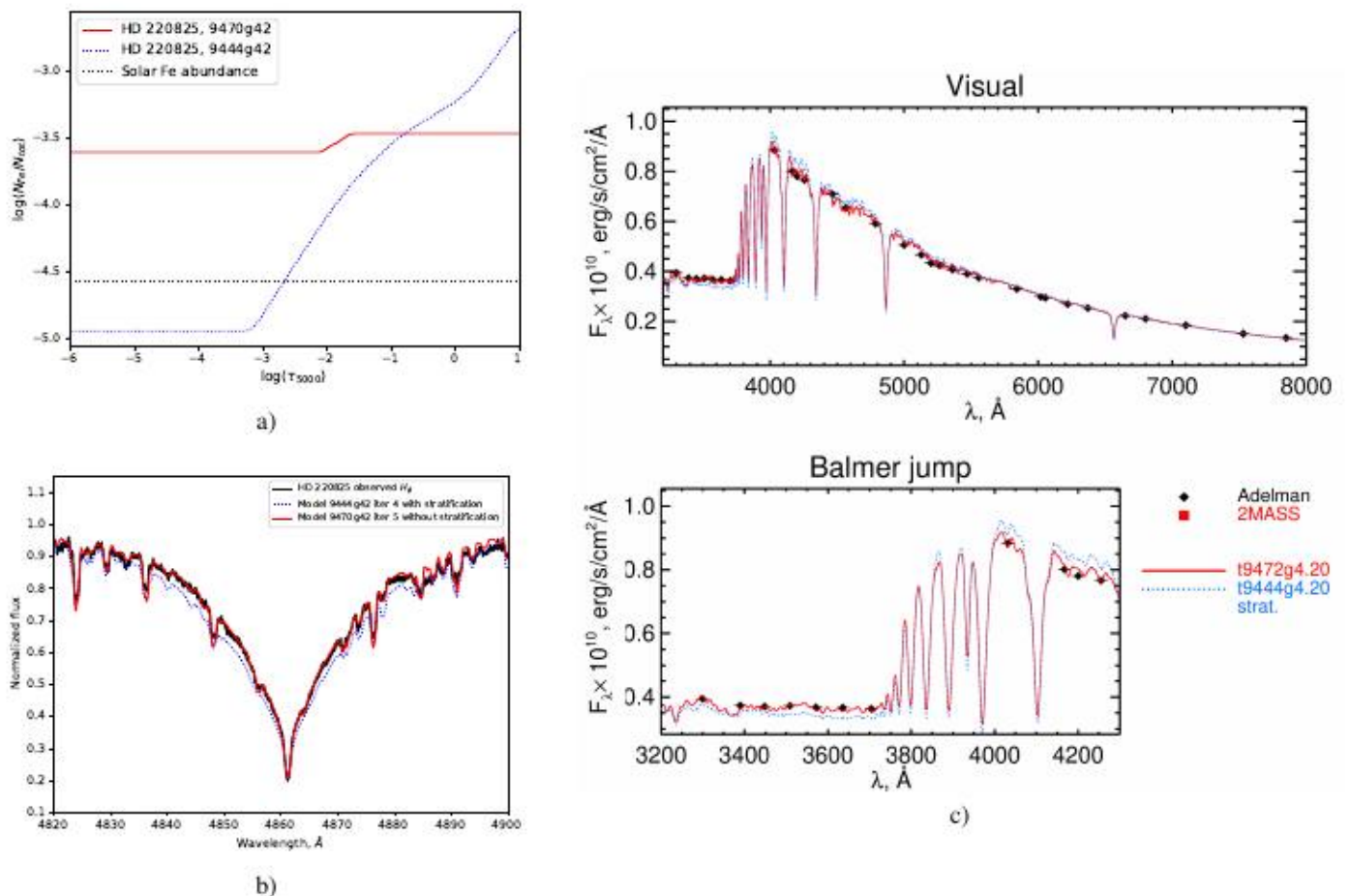


Fig. 3. a) – Vertical abundance distribution (stratification) of Fe in the atmosphere of HD 220825 derived for model 9470g42 (red line) and for model 9444g42 (blue dashed line). Solar abundance is denoted by black dotted line. b) – Comparison of observed H_β line profile in the atmosphere of HD 220825 (black line) with theoretical ones calculated assuming stratified (blue dashed line) and chemically homogeneous atmospheres (red line). c) – Comparison between the observed and best fit theoretical spectral energy distributions calculated with (blue dotted line) and without (solid red line) stratification. Observed SED data in visual spectral region and around the Balmer jump are shown with filled black diamonds.

dance pattern, we optimized T_{eff} , $\log g$, and stellar radii R to obtain their best fit values. The final fits are shown in Fig. 5 (HD 220825), Fig. 6 (HD 120198), and Fig. 7 (HD 4778), respectively. We found out that in all cases the values of $\log g$ could not be robustly determined from the fit to the observed energy distribution. Therefore we determined $\log g$ by fitting the hydrogen line profiles (see, for example, Fig. 4). For HD 4778 the value of $\log g = 4.0$ was taken, which resulted in the final best fit model of 9605g40. Note that for this star we were able to construct a model atmosphere even without spectrophotometric data in the optical region and using only Johnson's photometry.

Note that our final uncertainties on the stellar radii account for errors in both T_{eff} and parallax (distance) for all stars. The final fundamental parameters of the programme stars are collected in Table 5.

4.1. Effect of surface abundance inhomogeneity on the determination of fundamental parameters

The determination of fundamental parameters presented in our paper are based on fitting to the time-averaged spectrophotometric scan from Adelman's catalog. However, Ap stars are known to be photometric and spectroscopic variable due to the presence of abundance spots on their surfaces. These spots lead to the modulation of the SED as star rotates, and potentially cause

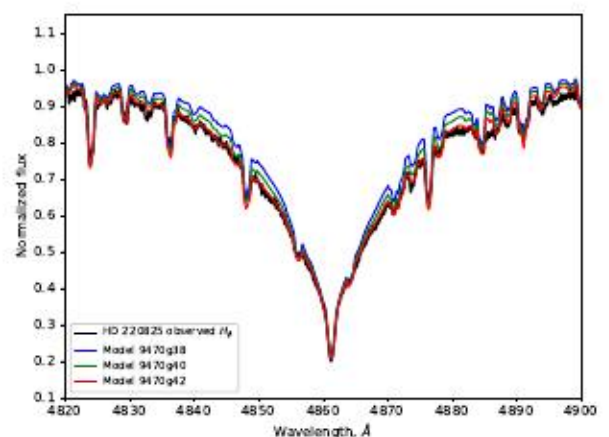


Fig. 4. Comparison between modelled and observed H_β line profile in HD 220825 for models with same $T_{\text{eff}} = 9470$ K and different $\log g = 3.8$ (blue dashed line), 4.0 (green dashed line) and 4.2 (red dashed line).

biases in the derived parameters such as, e.g., T_{eff} . To study the effects of surface inhomogeneity we investigated the case of HD 220825, a star which clearly shows a spotted surface (Ryabchikova et al. 1996) that can lead to light variability (see, for example, Shulyak et al. 2010a). Because we have spectrophoto-

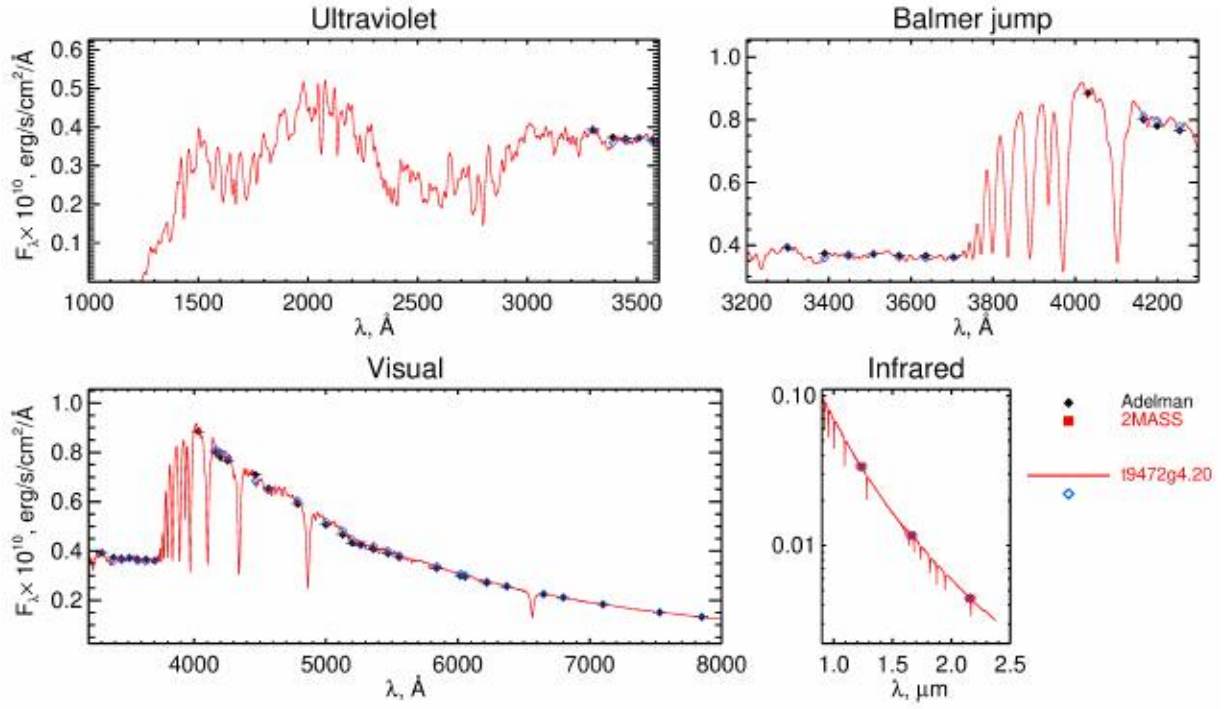


Fig. 5. Comparison between the observed energy distribution taken from Adelman’s (black filled diamonds) and 2MASS (filled red squares) catalogues and best fit theoretical flux (red solid line) calculated after five abundance analysis iterations for LLmodels atmosphere of HD 220825 with the parameters $T_{\text{eff}} = 9470$ K, $\log g = 4.2$. Open blue diamonds show the theoretical fluxes convolved with the corresponding filters that were used in actual observations.

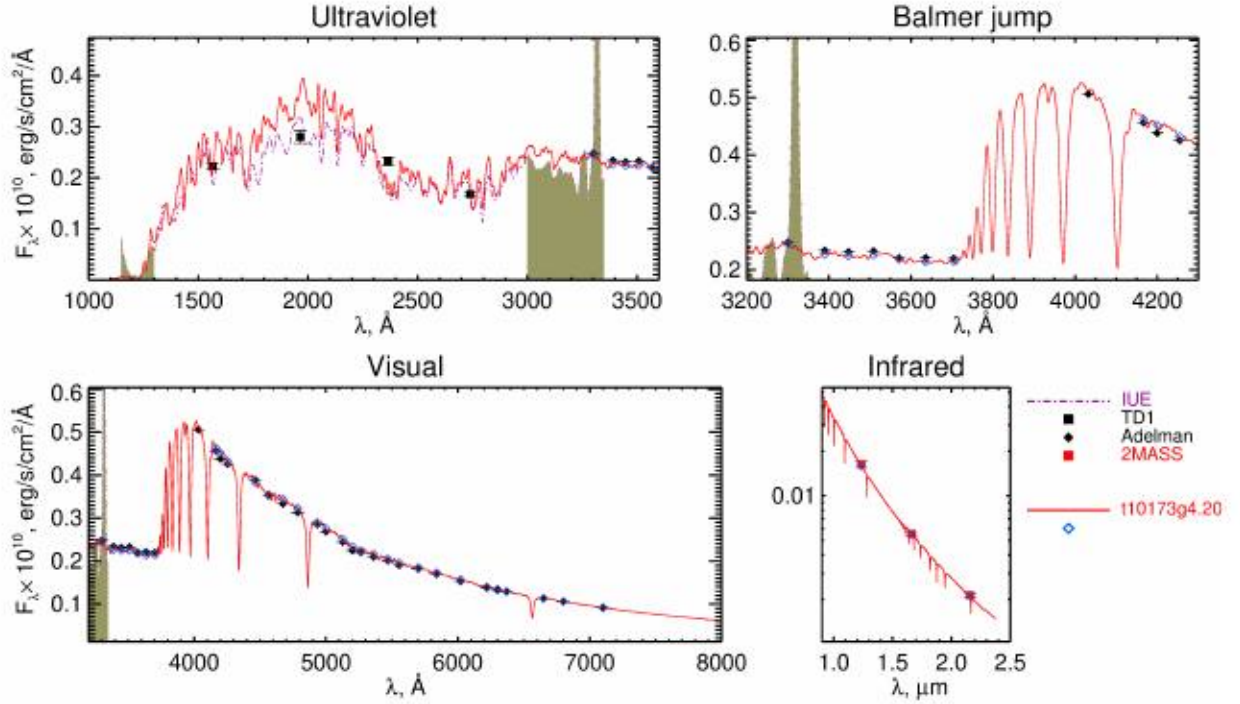


Fig. 6. Comparison of the observed energy distribution from Adelman’s (black filled diamonds), TD1 (black filled squares) and 2MASS photometric data (red filled squares) with the best fit theoretical flux (red solid line) calculated after five abundance analysis iterations for LLmodels atmosphere of HD 120198 with the parameters $T_{\text{eff}} = 10173$ K, $\log g = 4.2$. Open blue diamonds show the theoretical fluxes convolved with the corresponding filters. Shaded area illustrates spectral regions excluded from the fit.

tometric observations of Adelman obtained at rotation phases different from ESPaDOnS spectrum which we used for abundance analysis (rotation phase 0.011), it allowed us to estimate

possible errors in T_{eff} and $\log g$ associated with an inhomogeneous surface abundance distribution.

Spectrophotometric scans in 10 phases of rotation were extracted from the paper by Pyper & Adelman (1983). Table 4.1

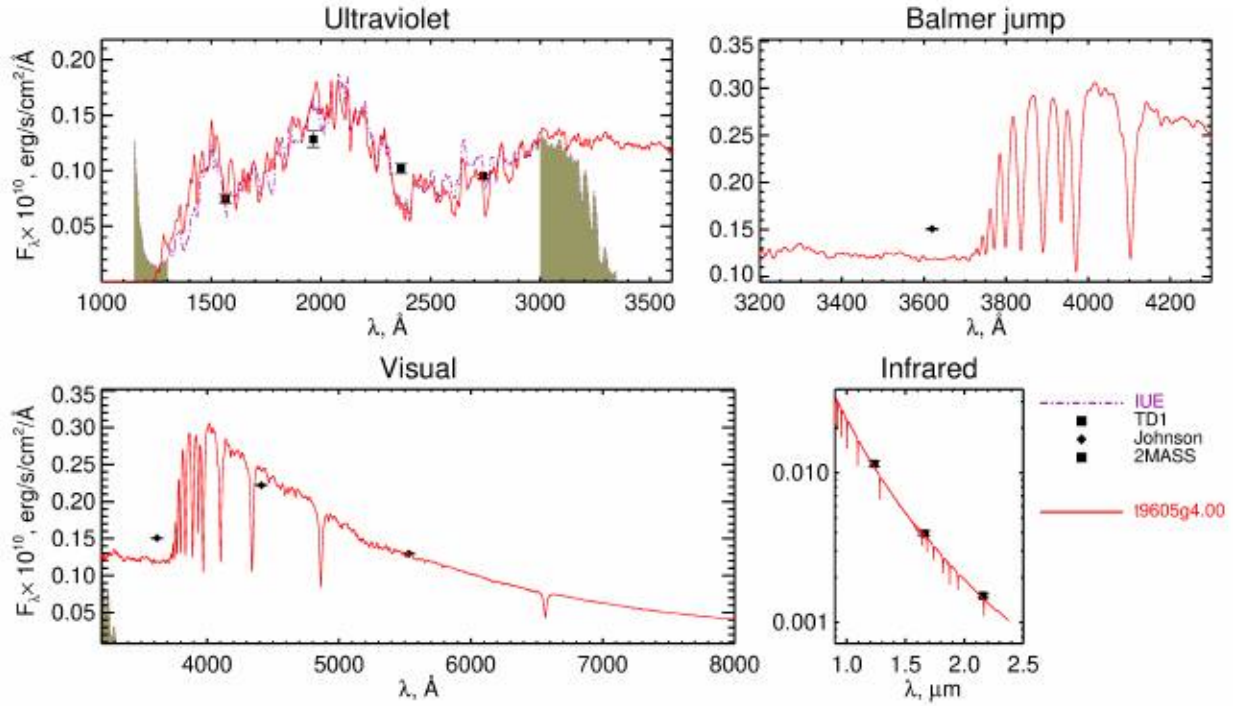


Fig. 7. Comparison of the observed energy distribution from Johnson (black filled diamonds), TD1 (black filled squares) and 2MASS photometry (black filled circles) with the best fit theoretical flux (red solid line) calculated after three abundance analysis iterations for LLMODELS atmosphere of HD 4778 with the parameters $T_{\text{eff}} = 9605$ K, $\log g = 4.0$. Shaded area illustrates spectral regions excluded from the fit.

provides T_{eff} values and their errors derived by fitting to these individual spectrophotometric scans. The $\log g$ was difficult to estimate unambiguously and we would simply set a rather conservative uncertainty of 0.2 dex ($\log g = 3.8, 4.0$, and 4.2 were used for choosing model atmosphere by fitting hydrogen line profiles).

Table 4. T_{eff} derived from fitting spectrophotometric scans obtained in 10 phases of rotation. The averaged T_{eff} with its standard deviation is given in the last row.

HJD	Phase	T_{eff} , K	Error
2441559.0	0.895	9415	94
2441561.0	0.385	9429	93
2441561.8	0.934	9545	37
2441563.0	0.703	9507	37
2441577.8	0.234	9488	34
2441578.8	0.942	9522	31
2441616.8	0.717	9451	25
2443061.8	0.587	9477	26
2443063.8	0.941	9498	23
2443066.8	0.985	9431	22
Average		9476	41

To estimate the effect of the flux variability caused by surface abundance spots we plotted averaged spectrophotometric scan (black filled circles) together with the scans obtained at different phases (coloured filled circles) in Fig. 8. We compared observations and the theoretical SED calculated with the finally adopted parameters for HD 220825: $T_{\text{eff}} = 9470$ and $\log g = 4.2$. In order to see the effect of variability on the determination of stellar parameters, we calculated the models with the parameters $T_{\text{eff}} = 9470 \pm 100$ K and $\log g = 3.8, 4.0$. One can see that the temperature variations within ± 100 K is enough to fit all observations. At the same time, changing $\log g$ has only little effect on SED. We thus conclude that the flux variations caused by the surface

inhomogeneity does not lead to effective temperature change by more than ± 100 K, which is a typical error in temperature estimates of Ap stars in similar self-consistent studies.

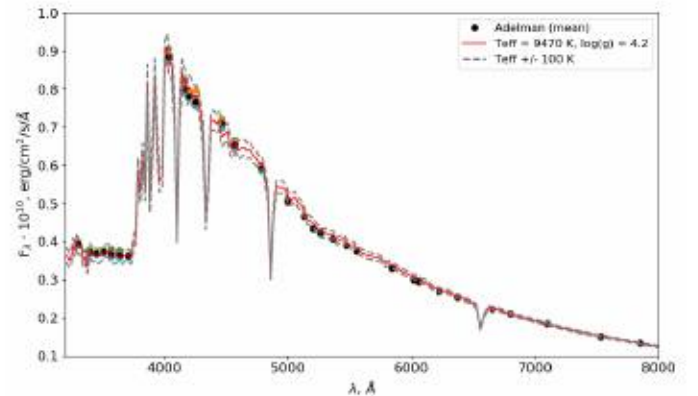


Fig. 8. Comparison between the observed Adelman's spectrophotometric scans (colour symbols) and theoretical spectral distribution for HD 220825. The solid red line shows the theoretical energy distribution computed for the final self-consistent model atmosphere of HD 220825 with parameters $T_{\text{eff}} = 9470$ K and $\log g = 4.2$. The grey dashed lines illustrate the effect of changing T_{eff} by ± 100 K. Averaged spectrophotometric scan is represented by filled black circles.

5. Comparison with the interferometry

The current study completes a programme of fundamental parameters determinations by means of spectroscopy of a sample of Ap stars that were chosen for interferometric measurements. The final sample of stars with fundamental parameters obtained from interferometric observations is given in paper by Perraut

et al. (2020). It contains 14 stars, and for all but one of them the self-consistent spectroscopic modelling was carried out. First, we compare spectroscopic and interferometric results obtained in this work.

HD 4778. The obtained $T_{\text{eff}}^{\text{interf}}$ is smaller than $T_{\text{eff}}^{\text{spec}}$. However note that the latter agrees well with photometric analysis within errors. We thus tried to fix the temperature by using additional diagnostic which is to fit $H\beta$ line profile with model atmospheres calculated assuming both T_{eff} estimates. The comparison is presented in Fig. 9a). For both temperatures the synthetic spectra were calculated with same abundances. It is seen that the model atmosphere with the spectroscopic temperature fits the $H\beta$ line profile noticeably better than the interferometric one.

HD 120198. While fundamental parameters of HD 120198 agree within the errors of determination, T_{eff} estimated by spectroscopy is higher and fits the observed $H\beta$ line better compared to the interferometric one (Fig. 9b).

HD 220825. For this star we obtained the largest difference between the effective temperatures derived via spectroscopy and interferometry. The difference exceeds the quoted errors. The radii of this star obtained with two methods are very close ($R/R_{\odot}^{\text{spec}} = 1.71$ and $R/R_{\odot}^{\text{interf}} = 1.78$), therefore the lower effective temperature derived based on interferometric analysis might be a result of the total flux underestimation by Perraut et al. (2020). There are no flux measurements of HD 220825 in the UV region, which forced Perraut et al. (2020) to estimate this flux by approximate correlations between flux in optical and UV deduced from the observations of other stars. We note that the spectroscopic temperature $T_{\text{eff}}=9470$ K fits $H\beta$ line profile much better than the interferometric one $T_{\text{eff}}=8790$ K, assuming same $\log g=4.2$ for both cases (see Fig. 9c). However, if we decrease the value of surface gravity from $\log g=4.2$ to $\log g=4.0$ then we obtain a reasonable fit to the observed $H\beta$ line profile with the interferometric temperature (Fig. 10, left panel). Note that fitting SED does not provide a robust estimate for $\log g$ in temperature range 9000 K–10000 K (see Section 4), and a decrease of $\log g$ does not noticeably change spectrophotometric fit, but a combination of the interferometric temperature with any of the $\log g$ -values does not fit the observed spectrophotometry, as demonstrated in Fig. 10 (right panel).

Table 5 collects fundamental parameters for 13 benchmarking stars obtained by interferometric (Perraut et al. 2020), photometric (Kochukhov & Bagnulo 2006), and spectroscopic methods. Parameters derived by indirect methods (spectroscopy and photometry) are plotted versus those derived by direct method (interferometry) in Fig. 11. Comparison of the parameters obtained by spectroscopy and interferometry shows a rather good agreement between them for all stars: the spectroscopic radii are on average by 5% smaller than the interferometric ones, which is within 2σ of the interferometric measurements. The luminosity agree even better. The fundamental parameters derived from photometric calibrations, on the whole, do not show significant systematic shifts relative to spectroscopic and, thus, interferometric ones. However, the dispersion in parameter's correlations is approximately two times larger than in spectroscopy. Our comparison shows that for stars with $T_{\text{eff}} > 9000$ K the spectroscopically derived temperatures are often significantly higher compared to interferometric estimates. This is the case for the mentioned above HD 220825 and HD 4778, and also for HD 108662 (see Table 5). The reason could be inaccurate radii determinations by interferometry. For instance, we notice that the difference between spectroscopic and interferometric determinations, on the average, is higher for stars with smaller an-

gular diameters $\theta < 0.3$ mas, which is very close to the current instrument limit. The largest difference is found for HD 108662, where the overestimation of stellar radii by interferometric measurements results in a T_{eff} which is more than 1300 K lower compared to the spectroscopic estimate. Or, as in the case of HD 220825, a lack of observed flux in the UV region where A-type stars radiate most of their energy could bias interferometric estimates towards lower T_{eff} values. To the contrary, a model fit to the observed SED does not strongly rely on the availability of observations in all spectral regions. It is mandatory though that all available flux observations are consistently fitted with a single model atmosphere, as we did in this study for HD 4778 where in the visual domain only two photometric flux measurements were available. The resulting T_{eff} is then set by the model input, and not by wavelength integration of the observed flux. Using model atmospheres also allows one to detect possible inconsistency between flux observed at different wavelength domains and instruments/missions, as was the case for, e.g., UV observations of HD 108662 (Romanovskaya et al. 2020b). The independent T_{eff} diagnostic – hydrogen Balmer lines – favors spectroscopically derived values for all these cases.

This comparison shows that spectroscopic methods can be successfully used to measure stellar radii with the sufficient accuracy for those Ap stars where interferometric observations are still impossible. It is also shown that a less time-consuming photometric method provides stellar parameters that are accurate enough to be used for statistical studies of a large sample of Ap stars.

Next, we used our spectroscopically derived parameters to place stars in the Hertzsprung-Russell (HR) diagram which we show on Fig. 12. The theoretical evolution tracks for solar metallicity models were taken from Girardi et al. (2000) in order to be consistent with Perraut et al. (2020). One can notice a trend of decreasing the strength of the magnetic field with age for stars with masses $> 2.5 M_{\odot}$. Three stars that are very close to their end on MS life (HD 204411, HD 40312, and HD 112185 marked as black circles on the plot) have weakest magnetic fields. However, the dependence of the magnetic field strength with age becomes less clear for stars with lower masses but the limited sample of stars does not allow us to make other conclusions about evolutionary changes of the magnetic fields. We only note that all three stars analysed in this work are relatively young but have magnetic fields smaller than the older stars in the same mass range. More work is needed to populate the HR diagram with stars analysed with our spectroscopic method in order to provide a consistent view on the evolution of the magnetic field in all mass ranges.

6. Conclusions

In this work we derived fundamental parameters of three Ap stars HD 4778, HD 120198, and HD 220825 thereby completing the detailed spectroscopic analysis of a sample of Ap stars for each of which independent estimates of their parameters were available from interferometric observations.

A rather high rotation velocity $v_e \sin i$ derived for all three stars complicated the determination of the chemical composition in their atmospheres. Therefore the abundances were obtained mainly through spectrum synthesis method by taking into account the contribution of numerous line blends. Chemical composition of our targets show a typical abundance pattern found in Ap stars: a deficiency of light elements and almost gradual increase of overabundance towards the heavy elements, reaching +3–4 dex overabundance in rare-earth element sequence. The

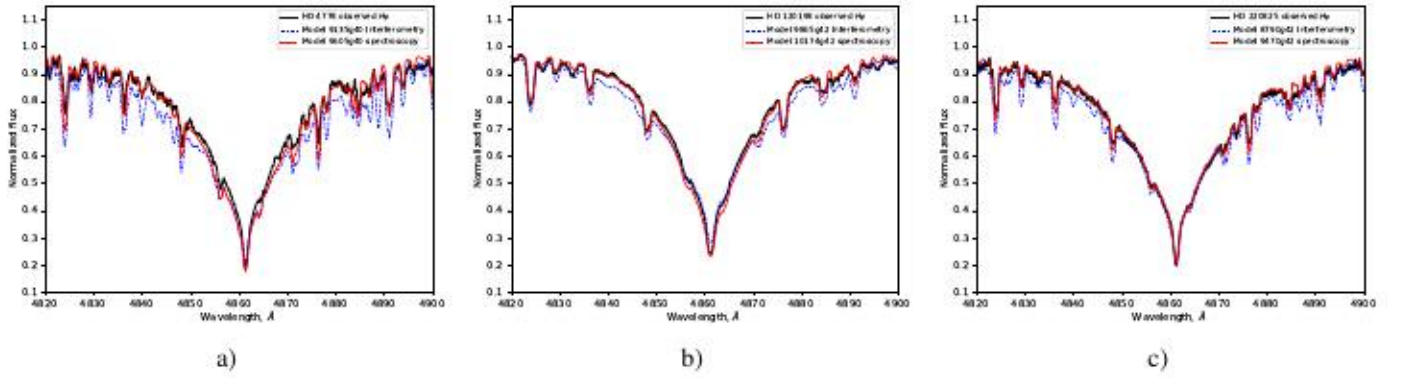


Fig. 9. Comparison between modelled and observed H_δ line profile in (a) HD 4778, (b) HD 120198, and (c) HD 220825 for models with same $\log g$ and different T_{eff} obtained by various methods: $T_{\text{eff}} = 9605$ K (HD 4778), 10174 K (HD 120198), 9470 K (HD 220825) by self-consistent modelling (red line) and $T_{\text{eff}} = 9135$ K (HD 4778), 9865 K (HD 120198), 8790 K (HD 220825) by interferometry (blue dashed line).

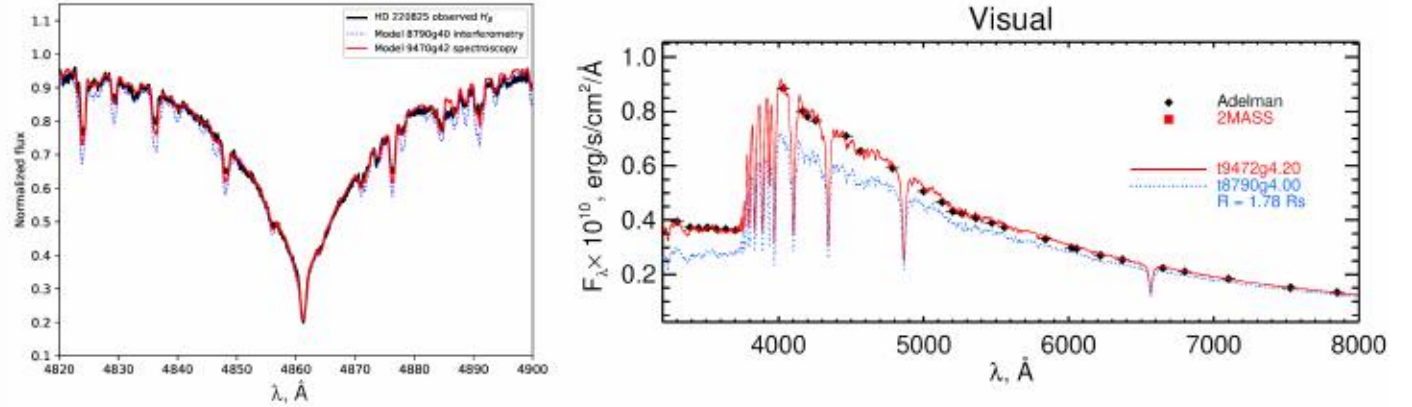


Fig. 10. Comparison between the observed H_δ line profile (left panel) and Adelman's spectrophotometric scan (right panel) and model fits calculated with spectroscopic (red lines) and interferometric T_{eff} of HD 220825 (blue dotted lines).

Table 5. Fundamental parameters of sample of Ap-stars derived with high resolution spectroscopy. In $\log g$ column symbol ":" means that the value was refined according to the hydrogen line profile.

Stars	Spectroscopy				Interferometry			Photometry			B_J , kGs
	T_{eff} , K	$\log g$	$R/R_\odot$	$\log L/L_\odot$	T_{eff} , K	$R/R_\odot$	$\log L/L_\odot$	T_{eff} , K	$R/R_\odot$	$\log L/L_\odot$	
4778	9605(92) ^(a)	4.00:	2.14(4)	1.54(3)	9135(400)	2.36(12)	1.54(5)	9977(300)	1.91(4)	1.51(7)	2.6
24712	7250(100) ^(b)	4.10(15)	1.77(4)	0.89(7)	7235(280)	1.75(5)	0.88(7)	7194(200)	1.76(4)	0.87(4)	2.3
108662	10212(100) ^(c)	4.20:	2.09(10)	1.63(6)	8880(330)	2.59(12)	1.58(6)	10495(400)	2.43(1)	1.81(7)	3.3
118022	9140(30) ^(d)	4.20(2)	2.16(6)	1.47(3)	9100(190)	2.17(6)	1.46(5)	9057(290)	2.18(4)	1.46(4)	3.0
120198	10174(100) ^(a)	4.20:	2.18(4)	1.66(3)	9865(370)	2.28(10)	1.65(4)	10543(390)	2.01(1)	1.65(6)	1.6
128898	7500(130) ^(e)	4.10(15)	1.94(1)	1.03(3)	7420(170)	1.97(6)	1.02(2)	7673(190)	1.88(7)	1.04(1)	2.0
137909	8100(150) ^(f)	4.00(15)	2.47(7)	1.37(8)	7980(180)	2.63(9)	1.40(5)	7430(200)	3.17(9)	1.44(2)	5.4
153882	9150(100) ^(g)	4.01(13)	3.35(3)	1.85(3)	8980(600)	3.46(37)	1.85(4)	9727(290)	3.53(23)	2.00(11)	3.8
176232	7550(50) ^(h)	3.80(10)	2.46(6)	1.29(4)	7900(190)	2.21(8)	1.23(4)	7925(200)	2.43(1)	1.32(4)	1.5
188041	8770(150) ⁽ⁱ⁾	4.20(10)	2.39(7)	1.48(3)	9000(360)	2.26(5)	1.48(6)	8433(200)	2.80(9)	1.55(7)	3.6
201601	7550(150) ^(f)	4.00(10)	2.07(5)	1.10(7)	7253(235)	2.11(7)	1.04(4)	7620(200)	2.04(3)	1.10(3)	4.0
204411	8300(150) ⁽ⁱ⁾	3.60(10)	4.42(15)	1.92(6)	8520(220)	4.23(11)	1.93(5)	8749(300)	4.12(1)	1.95(6)	0.8
220825	9470(100) ^(a)	4.20:	1.71(4)	1.32(4)	8790(230)	1.78(3)	1.24(4)	9078(290)	1.92(3)	1.35(4)	2.0

Note: ^(a) Current work, ^(b) Shulyak et al. (2009), ^(c) Romanovskaya et al. (2020b), ^(d) Kochukhov et al. (2019), ^(e) Kochukhov et al. (2009), ^(f) Shulyak et al. (2013), ^(g) Romanovskaya et al. (2020a), ^(h) Nesvacil et al. (2013), ⁽ⁱ⁾ Romanovskaya et al. (2019).

only exception is Ba which abundance drops to the solar value. This is opposite to Ba abundance behaviour in superficially-normal A and Am stars where overabundances of heavy elements are also observed, and Ba is the most overabundant element in Sr-Y-Zr-Ba-Nd sequence (Mashonkina et al. 2020). It may indicate that different mechanisms produce overabundance of heavy elements in the atmospheres of different groups of A-stars.

We did not find significant Fe and Cr stratification typical for cooler Ap atmospheres. It might be caused by a difficulty in sampling the suitable lines for stratification analysis due to rather high rotation velocities. Therefore the final modelling was performed assuming chemically homogeneous atmospheres.

Fundamental parameters derived by self-consistent spectroscopic modelling that take into account the anomalous chemical composition of stellar atmospheres were compared with the parameters obtained by direct interferometric measurements. We

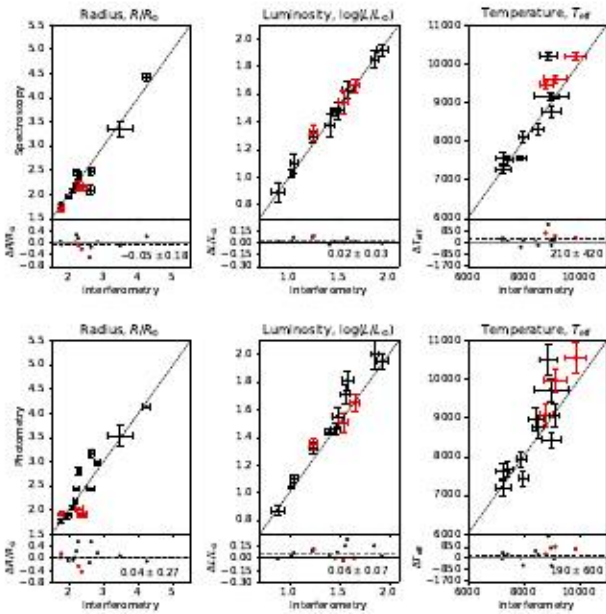


Fig. 11. Comparison between radii, luminosity, and effective temperatures obtained by spectroscopy, photometry, and interferometry.

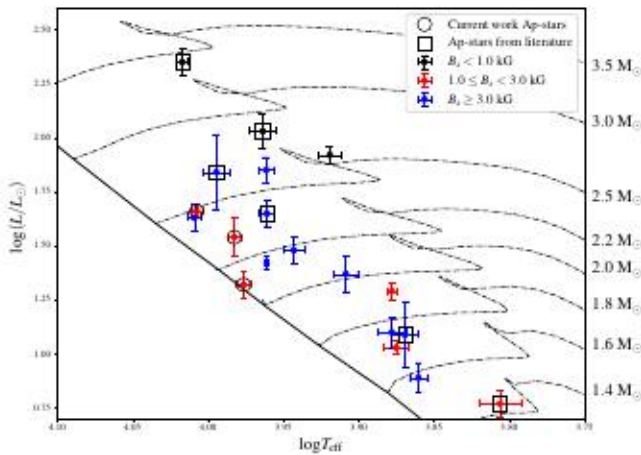


Fig. 12. Ap stars on HR-diagram. Position of stars from this work are marked with open circles. Six stars that were not part of the interferometric sample are marked with open squares. The symbols are color coded according to the surface magnetic field strength (see figure legend).

found that stellar radii and luminosity agree within the errors, while spectroscopically determined effective temperatures are higher than those derived by interferometry for hotter stars with $T_{\text{eff}} > 9000$ K. The possible reason of such a discrepancy is the underestimation of UV contribution to the total flux balance in the interferometric studies, and/or intrinsic inaccuracy in radii measurements for stars with small angular diameters. In case of HD 4778, we succeeded to construct a model atmosphere when only two points of Johnson's photometry were available in the optical region.

Using HD 220825 as an example, we estimated a possible influence of the surface chemical inhomogeneity on the derived fundamental parameters through modelling of the observed

spectral energy distribution at different phases of stellar rotation. We found out that surface flux variability translates to ± 100 K effective temperature variations which is a typical error for Ap stars and thus can be neglected in the determination of fundamental parameters in future works.

Acknowledgements. This research has made use of the data from GAIA DR2 catalogues through the VizieR catalogue access tool. The use of the VALD database is acknowledged. This work is based on observations obtained with ESPODOnS spectropolarimeter. We use the archived data from ESPODOnS spectrograph with programme ID: ID 09BQ78 (HD 4778), ID 16AC27 (HD 120198), and ID 18BC22 (HD 220825). A. Romanovskaya acknowledges the financial support from the grant RFBR, project number 19-32-90147. D. Shulyak acknowledges the financial support from the State Agency for Research of the Spanish MCIU through the "Center of Excellence Severo Ochoa" award to the Instituto de Astrofísica de Andalucía (SEV-2017-0709).

References

- Adelman, S. J., Pyper, D. M., Shore, S. N., White, R. E., & Warren, Jr., W. H. 1989, *A&AS*, 81, 221
- Adelman, S. J. & Rayle, K. E. 2000, *A&A*, 355, 308
- Alecian, G. & Stift, M. J. 2010, *A&A*, 516, A53
- Amôres, E. B. & Lépine, J. R. D. 2005, *AJ*, 130, 659
- Asplund, M., Amarsi, A. M., & Grevesse, N. 2021, *arXiv e-prints*, arXiv:2105.01661
- Aurière, M., Wade, G. A., Silvester, J., et al. 2007, *A&A*, 475, 1053
- Babel, J. 1992, *A&A*, 258, 449
- Cohen, M., Wheaton, W. A., & Megeath, S. T. 2003, *AJ*, 126, 1090
- Cutri, R. M., Skrutskie, M. F., van Dyk, S., et al. 2003, *VizieR Online Data Catalog*, 2246
- Gaia Collaboration. 2018, *VizieR Online Data Catalog*, I/345
- Girardi, L., Bressan, A., Bertelli, G., & Chiosi, C. 2000, *A&AS*, 141, 371
- Glagolevskij, Y. V. 2019, *Astrophysical Bulletin*, 74, 66
- Hubrig, S., North, P., & Schöller, M. 2007, *Astronomische Nachrichten*, 328, 475
- Kochukhov, O. 2017, *A&A*, 597, A58
- Kochukhov, O. 2018, *BinMag: Widget for comparing stellar observed with theoretical spectra*, *Astrophysics Source Code Library*
- Kochukhov, O. & Bagnulo, S. 2006, *A&A*, 450, 763
- Kochukhov, O., Makaganiuk, V., & Piskunov, N. 2010, *A&A*, 524, A5
- Kochukhov, O., Shultz, M., & Neiner, C. 2019, *A&A*, 621, A47
- Kochukhov, O., Shulyak, D., & Ryabchikova, T. 2009, *A&A*, 499, 851
- Kochukhov, O. P. 2007, in *Physics of Magnetic Stars*, ed. I. I. Romanyuk, D. O. Kudryavtsev, O. M. Neizvestnaya, & V. M. Shapoval, 109–118
- Krtićka, J., Mikulášek, Z., Lüftinger, T., et al. 2012, *A&A*, 537, A14
- Kupka, F., Piskunov, N., Ryabchikova, T. A., Stempels, H. C., & Weiss, W. W. 1999, *A&AS*, 138, 119
- Kurucz, R. 1993, *ATLAS9 Stellar Atmosphere Programs and 2 km/s grid*. Kurucz CD-ROM No. 13. Cambridge, Mass.: Smithsonian Astrophysical Observatory, 1993., 13
- Lallement, R., Vergely, J. L., Valette, B., et al. 2014, *A&A*, 561, A91
- Landi Degl'Innocenti, E. 1975, *A&A*, 45, 269
- LeBlanc, F., Monin, D., Hui-Bon-Hoa, A., & Hauschildt, P. H. 2009, *A&A*, 495, 937
- Lipski, E. & Stepień, K. 2008, *MNRAS*, 385, 481
- Mashonkina, L., Ryabchikova, T., Alexeeva, S., Sitnova, T., & Zatsarinny, O. 2020, *MNRAS*, 499, 3706
- Mathys, G. 2017, *A&A*, 601, A14
- Michaud, G. 1970, *ApJ*, 160, 641
- Nesvacił, N., Shulyak, D., Ryabchikova, T. A., et al. 2013, *A&A*, 552, A28
- Netopil, M., Paunzen, E., Maitzen, H. M., North, P., & Hubrig, S. 2008, *A&A*, 491, 545
- Pandey, C. P., Shulyak, D. V., Ryabchikova, T., & Kochukhov, O. 2011, *MNRAS*, 417, 444
- Perraut, K., Cunha, M., Romanovskaya, A., et al. 2020, *A&A*, 642, A101
- Przybilla, N., Butler, K., Becker, S. R., Kudritzki, R. P., & Venn, K. A. 2000, *A&A*, 359, 1085
- Pyper, D. M. & Adelman, S. J. 1983, *A&AS*, 51, 365
- Romanovskaya, A., Ryabchikova, T., Shulyak, D., et al. 2019, *MNRAS*, 488, 2343
- Romanovskaya, A. M., Ryabchikova, T. A., & Shulyak, D. V. 2020a, *Astronomy Letters*, 46, 331
- Romanovskaya, A. M., Ryabchikova, T. A., & Shulyak, D. V. 2020b, *INASAN Science Reports*, 5, 219
- Romanyuk, I. I. & Kudryavtsev, D. O. 2008, *Astrophysical Bulletin*, 63, 139
- Ryabchikova, T., Leone, F., & Kochukhov, O. 2005, *A&A*, 438, 973

- Ryabchikova, T., Piskunov, N., Kurucz, R. L., et al. 2015, *Phys. Scr.*, 90, 054005
- Ryabchikova, T. A., Pavlova, V. M., Davydova, E. S., & Piskunov, N. E. 1996, *Astronomy Letters*, 22, 822
- Ryabchikova, T. A. & Romanovskaya, A. M. 2017, *Astronomy Letters*, 43, 252
- Shulyak, D., Krtićka, J., Mikulášek, Z., Kochukhov, O., & Lüftinger, T. 2010a, *A&A*, 524, A66
- Shulyak, D., Ryabchikova, T., Kildiyarova, R., & Kochukhov, O. 2010b, *A&A*, 520, A88
- Shulyak, D., Ryabchikova, T., & Kochukhov, O. 2013, *A&A*, 551, A14
- Shulyak, D., Ryabchikova, T., Mashonkina, L., & Kochukhov, O. 2009, *A&A*, 499, 879
- Shulyak, D., Tsymbal, V., Ryabchikova, T., Stütz, C., & Weiss, W. W. 2004, *A&A*, 428, 993
- Sikora, J., Wade, G. A., Power, J., & Neiner, C. 2019, *MNRAS*, 483, 2300
- Sitnova, T. M. & Mashonkina, L. I. 2018, *Astronomy Letters*, 44, 411
- Sitnova, T. M., Mashonkina, L. I., & Ryabchikova, T. A. 2013, *Astronomy Letters*, 39, 126
- ten Brummelaar, T. A., McAlister, H. A., Ridgway, S. T., et al. 2010, in *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, Vol. 7734, *Optical and Infrared Interferometry II*, ed. W. C. Danchi, F. Delplancke, & J. K. Rajagopal, 773403
- Thompson, G. I., Nandy, K., Jamar, C., et al. 1978, *Catalogue of stellar ultraviolet fluxes. A compilation of absolute stellar fluxes measured by the Sky Survey Telescope (S2/68) aboard the ESRO satellite TD-1*
- van Leeuwen, F. 2007, *A&A*, 474, 653
- Wade, G. A., Ryabchikova, T. A., Bagnulo, S., & Piskunov, N. 2001, in *Astronomical Society of the Pacific Conference Series*, Vol. 248, *Magnetic Fields Across the Hertzsprung-Russell Diagram*, ed. G. Mathys, S. K. Solanki, & D. T. Wickramasinghe, 373